CognitiveOS vs. Doot / Agent One — Compliance

Technical / architectural alignment against the requirements described in the Doot white paper

Paper requirement	CognitiveOS	Status
Personal / individual agent	Actor-specific CognitiveOS	95%
Actor-local cognition	Private beliefs, memory, goals, execution	100%
Identity-bound authority	Identity / authorization / delegation	~90%
Consent & least privilege	Consent, scopes, revocation	~90%
Secure tool execution	Capability security + policy + TransitionGate	~95%
Human approval for high-risk actions	Human approval / WAITING_FOR_HUMAN	100%
Policy-bound reasoning / execution	Policy engine / governance	~95%
Privacy / private cognition	Actor-isolated cognition	~95%
Auditability / forensics	Audit + execution provenance	~95%
Agent isolation	One CognitiveOS per actor	100% architecturally
Agent-to-agent coordination	AgentBus / delegation / negotiation	~95%
Negotiated multi-agent actions	NegotiationPlanner + TransactionCoordinator	~95%
Shared world / state	Shared World + actor-local state	~95%
Dynamic planning	World-grounded dynamic ExecutionGraph	100%
Replanning after world change	Observe → Compare → Learn → Replan	~100%
Capability-scoped execution	CapabilityRegistry / ProviderRegistry	~95%
Provider interoperability	Python / REST / n8n / OpenClaw / ARD	~95%
LLM-agnostic architecture	Optional LLM planning + provider separation	~95%
Edge-first architecture	Per-actor edge deployment	~90%
Cloud / edge separation	Edge/Cloud architecture + sync	~90%
Resilience / failure isolation	Actor isolation + recovery	~90%
Observability	Lemon / execution / security monitoring	~95%
Contract-bound capabilities	Capability security / policy	~90%
Specification before intelligence	Deterministic runtime + policy gates	~95%
Blast-radius control	Actor/capability/policy boundaries	~90%
Failure containment	Isolation + recovery mechanisms	~85–90%
Rogue-agent deplatforming	Revocation / membership controls	~80–85%
Agent registry / control plane	Actor/Agent registries	~90%
Capability revocation	Revocation/security layer	~90%
Open standards	Provider/capability abstractions	~75–80%
Citizen-facing voice / multilingual UX	Not CognitiveOS core	Not implemented
Aadhaar / ABHA integration	Not implemented	Not implemented
DigiLocker integration	Not implemented	Not implemented
UPI integration	Not implemented as DPI infrastructure	Not implemented
Personhood Credentials / PHC	Not implemented	Not implemented
TEE / remote attestation	Not implemented	Not implemented

Paper requirement	CognitiveOS	Status
National-scale DPI deployment	Architecture supports scaling, not proven	Not qualified
22-language citizen experience	Not a current CognitiveOS feature	Not implemented
National regulatory framework	Outside product scope	N/A
National dispute-resolution infrastructure	Outside product scope	N/A

Overall: ~78–82% full Agent One / DPI implementation alignment; ~90%+ technical architecture alignment; ~94–95% CognitiveOS-specific completion. Main gaps: DPI identity rails, TEE/attestation, national-scale qualification, citizen UX/language requirements, and national governance infrastructure.